

# **HTML(HYPER TEXT MARKUP LANGUAGE)**

- HTML is the standard markup language for creating Web pages.
- HTML stands for Hyper Text Markup Language
- HTML is the standard markup language for creating Web pages
- HTML describes the structure of a Web page
- HTML consists of a series of elements
- HTML elements tell the browser how to display the content
- HTML elements label pieces of content such as "this is a heading", "this is a paragraph", "this is a link", etc.

# HTML History

| Year | Version                                                 |
|------|---------------------------------------------------------|
| 1989 | Tim Berners-Lee invented www                            |
| 1991 | Tim Berners-Lee invented HTML                           |
| 1993 | Dave Raggett drafted HTML+                              |
| 1995 | HTML Working Group defined HTML 2.0                     |
| 1997 | W3C Recommendation: HTML 3.2                            |
| 1999 | W3C Recommendation: HTML 4.01                           |
| 2000 | W3C Recommendation: XHTML 1.0                           |
| 2008 | WHATWG HTML5 First Public Draft                         |
| 2012 | <a href="#">WHATWG HTML5 Living Standard</a>            |
| 2014 | <a href="#">W3C Recommendation: HTML5</a>               |
| 2016 | W3C Candidate Recommendation: HTML 5.1                  |
| 2017 | <a href="#">W3C Recommendation: HTML5.1 2nd Edition</a> |
| 2017 | <a href="#">W3C Recommendation: HTML5.2</a>             |

# A Simple HTML Document

## Example:

### Code:

```
<!DOCTYPE html>
<html>
<head>
<title>Page Title</title>
</head>
<body>
<h1>My First Heading</h1>
<p>My first paragraph.</p>
</body>
</html>
```

### Output:

**My First Heading**

My first paragraph.

- The `<!DOCTYPE html>` declaration defines that this document is an HTML5 document
- The `<html>` element is the root element of an HTML page
- The `<head>` element contains meta information about the HTML page
- The `<title>` element specifies a title for the HTML page (which is shown in the browser's title bar or in the page's tab)
- The `<body>` element defines the document's body, and is a container for all the visible contents, such as headings, paragraphs, images, hyperlinks, tables, lists, etc.
- The `<h1>` element defines a large heading
- The `<p>` element defines a paragraph

# What is HTML5

- HTML5 is the fifth version of Hypertext Markup Language (HTML), a standard language used to structure webpages. It defines how content on a webpage should be structured and displayed. Here are some key points of HTML5.
- Provides multimedia support by embedding audio and video without plugins.
- Adds new form controls like date and email input types.
- Enables web storage to store data offline for better performance.
- Uses semantic elements such as <header> and <footer> for clearer structure.
- Delivers improved performance, especially on mobile devices.

# <html> Element

- The **lang** attribute specifies the **language** of the HTML document.
- It is added to the <**html**> tag.
- It helps **browsers** display the content correctly.
- It helps **search engines** understand the language of the webpage.
- It improves **accessibility** for screen readers and assistive technologies.
- Different language codes can be used, such as:
  - en → English
  - fr → French
  - es → Spanish
  - hi → Hindi
  - pa → Punjabi

<**html lang="en">**

# <head> element

## Code:

```
<!DOCTYPE html>
<html>
  <head>
    <title>My Web Page</title>
  </head>
  <body>
    <h1>Welcome to my web page!</h1>
    <p>This is the content of my web page.</p>
  </body>
</html>
```

## Output:

**Welcome to my web page!**

This is the content of my web page.

the <head> element contains several elements that provide metadata about the document:

- <title>: This element specifies the title of the web page, which appears in the browser's title bar and is used by search engines to index the page.
- <meta>: This element provides information about the character set used in the document. In this example, the charset attribute specifies that the document is encoded using UTF-8.
- <link>: This element specifies the location of an external stylesheet that is used to define the visual style of the web page.

# <body> element

- The <body> element contains all the **visible content** of a webpage.
- It is the **main container** for the content displayed in a web browser.
- It appears **after the <head> element** in an HTML document.
- The <head> element stores **metadata, title, links**, which are **not directly visible** on the webpage.
- The browser renders everything inside the <body> tag.
- The <body> can contain:
  - Headings (<h1> to <h6>)
  - Paragraphs (<p>)
  - Images (<img>)
  - Videos (<video>)
  - Links (<a>)
  - Lists (<ul>, <ol>)
  - Tables (<table>)
  - Forms (<form>)
  - Buttons (<button>)
  - Other HTML elements

# <body> element

## Code:

```
<!DOCTYPE html>
<html>
  <head>
    <title>My Webpage</title>
  </head>
  <body>
    <h1>Welcome to My Webpage</h1>
    <p>This is some sample text that will be displayed in the browser window.</p>
    
  </body>
</html>
```

## Output:

**Welcome to My Webpage**

This is some sample text that will be displayed in the browser window.

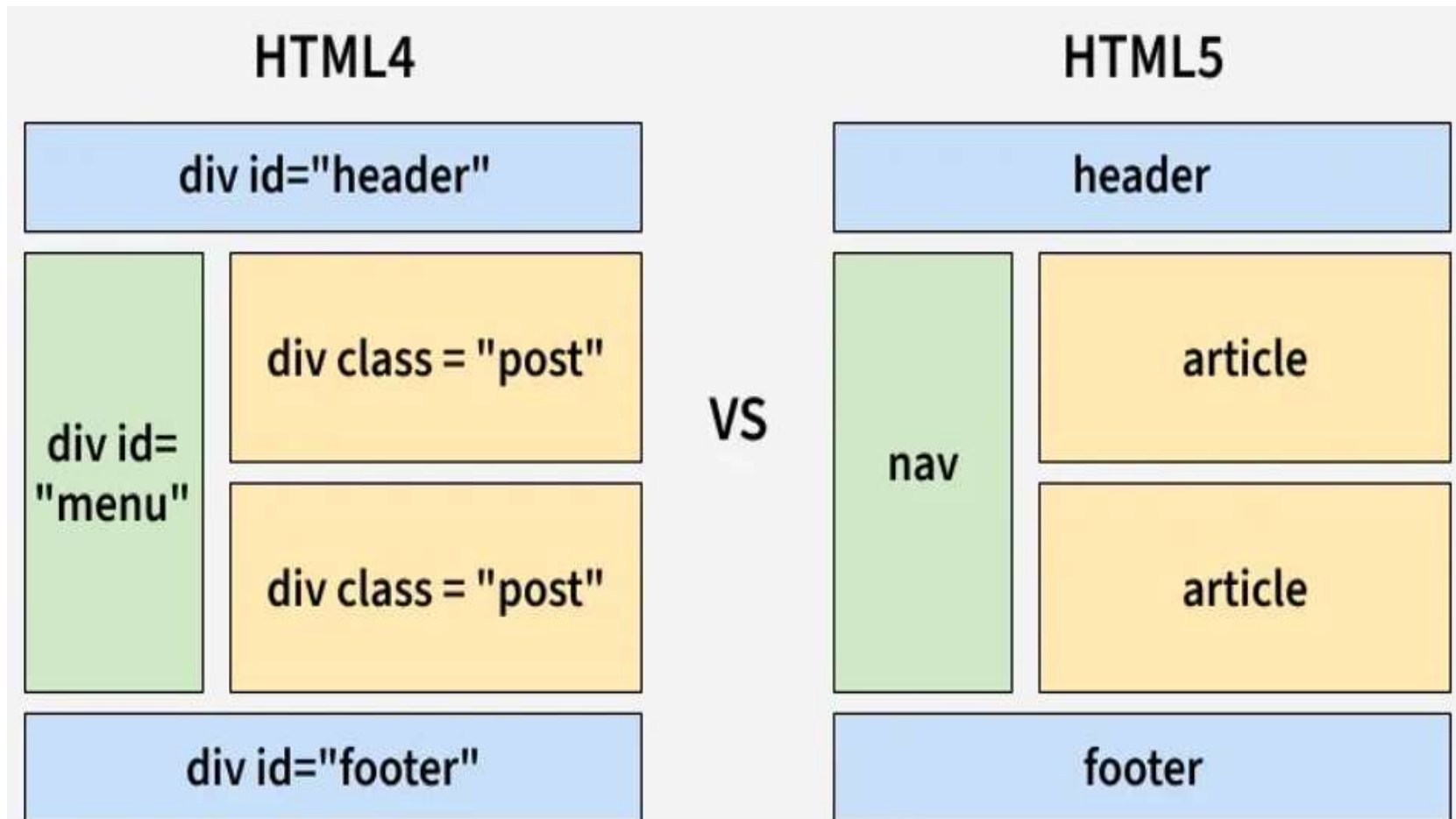
 My Image

# The Opening and the Closing Tags

- An HTML element is usually made up of:
  - **Opening (Start) Tag**
  - **Content**
  - **Closing (End) Tag**
- The **opening tag** starts with `<` and ends with `>`.
- The **opening tag** contains the **name of the HTML element**.
- The **content** is the information displayed between the opening and closing tags.
- The **closing tag** is similar to the opening tag but includes a **forward slash** (`/`) before the element name.

**`<p>This is a paragraph.</p>`**

# HTML4 VS HTML5



# Example of HTML5

**Code:**

```
<html>
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Simple HTML Page</title>
</head>
<body>
  <header>
    <h1>Hello, World!</h1>
  </header>
  <main>
    <p>This is a simple HTML5 page.</p>
  </main>
  <footer>
    <p>Chitkara University</p>
  </footer>
</body>
</html>
```

**Output:**

**Hello, World!**

This is a simple HTML5 page.

Chitkara University

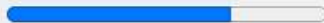
## Using the <progress> Element to Show Task Progress

### Code:

```
<html>
<body>
  <h1>HTML5 Progress Bar Example</h1>
  <progress value="70" max="100"></progress>
</body>
</html>
```

### Output:

## HTML5 Progress Bar Example



# Features of HTML5



<Audio>



<3D>



<Video>

New Features of HTML5



<Games>



<Storage>



<Type>



<Motion>

# New Added Elements in HTML 5

- **<article>**: The [<article>](#) tag represents independent, self-contained content like articles or posts.
- **<aside>**: The [<aside>](#) tag holds secondary or related information supporting the main content.
- **<figcaption>**: The [<figcaption>](#) tag provides a caption or description for a <figure> element.
- **<figure>**: The [<figure>](#) tag groups self-contained media such as images, diagrams, or code.
- **<header>**: The [<header>](#) tag contains introductory content like headings or navigation links.
- **<footer>**: The [<footer>](#) tag includes closing content such as author info or copyright notes.
- **<main>**: The [<main>](#) tag defines the central, most important content of the webpage.
- **<mark>**: The [<mark>](#) tag highlights or emphasizes specific text.
- **<nav>**: The [<nav>](#) tag groups navigation links for moving around the site.
- **<section>**: The [<section>](#) tag organizes related content into thematic blocks.

# New Added Elements in HTML 5

- **<details>**: The [<details>](#) tag creates expandable or collapsible content sections.
- **<summary>**: The [<summary>](#) tag provides the clickable heading for a <details> element.
- **<time>**: The [<time>](#) tag represents dates or times in a readable and machine-usable format.
- **<bdi>**: The [<bdi>](#) tag isolates text with unknown or mixed text direction.
- **<wbr>**: The [<wbr>](#) tag adds an optional line-break position in long words.
- **<datalist>**: The [<datalist>](#) tag provides autocomplete suggestions for an <input> element.
- **<keygen>**: The [<keygen>](#) tag generates key pairs for secure authentication (deprecated).
- **<output>**: The [<output>](#) tag displays results of calculations or script actions.
- **<progress>**: The [<progress>](#) tag visually represents the progress of a task.
- **<svg>**: The [<svg>](#) tag embeds scalable vector graphics into a webpage.

- **<audio>**: The [<audio>](#) tag embeds audio content like music or sound effects.
- **<embed>**: The [<embed>](#) tag inserts external resources such as media players or apps.
- **<source>**: The [<source>](#) tag defines media sources for <audio> and <video> elements.
- **<track>**: The [<track>](#) tag provides captions or subtitles for videos.
- **<video>**: The [<video>](#) tag embeds video content directly into a webpage.

# HTML5 Advantages

- HTML5 offers several benefits that enhance modern web development like:
- Works on all major browsers and devices (cross-platform).
- Lets websites access user location (geolocation).
- Supports offline use through Web Storage or AppCache.
- Uses semantic tags for cleaner, simpler code.
- Optimized for smooth performance on mobile devices.

# HTML5 Disadvantages

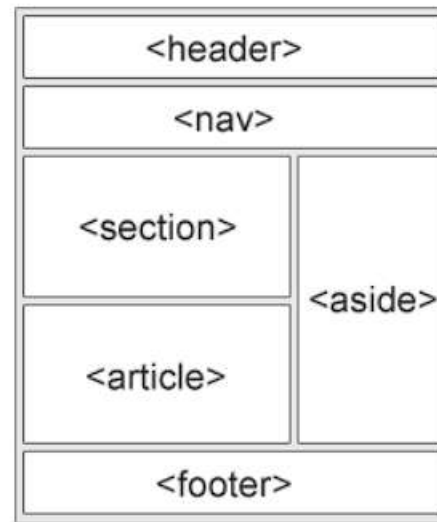
- Despite its strengths, HTML5 comes with some drawbacks:
- Older browsers may not support all HTML5 features (browser inconsistencies).
- Some older mobile devices struggle with HTML5 (mobile compatibility issues).
- Web Storage and similar features can pose security risks if misused.
- Advanced APIs introduce complexity for beginners.
- IE 8 and below show poor support for HTML5.

# What are Semantic Elements?

- A semantic element clearly describes its meaning to both the browser and the developer.
- Examples of **non-semantic** elements: `<div>` and `<span>` - Tells nothing about its content.
- Examples of **semantic** elements: `<img>`, `<table>`, and `<article>` - Clearly defines its content.

# Semantic Elements in HTML

- `<header>`
- `<nav>`
- `<section>`
- `<article>`
- `<aside>`
- `<footer>`



# HTML <header> Element

- The <header> element represents a container for introductory content or a set of navigational links.
- A <header> element typically contains:
  - one or more heading elements (<h1> - <h6>)
  - logo or icon
  - authorship information

**Note:** You can have several <header> elements in one HTML document. However, <header> cannot be placed within a <footer>, <address> or another <header> element.

# Example

## Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Header Example</title>
</head>
<body>

  <article>
    <header>
      <h1>What Does WWF Do?</h1>
      <p>WWF's mission:</p>
    </header>

    <p>
      WWF's mission is to stop the degradation of
our planet's natural
      environment, and build a future in which
humans live in harmony
      with nature.
    </p>
  </article>

</body>
</html>
```

## Output:

### What Does WWF Do?

WWF's mission:

WWF's mission is to stop the degradation of our planet's natural environment, and build a future in which humans live in harmony with nature.

# HTML <nav> Element

The <nav> element defines a set of navigation links.

## Code

```
<!DOCTYPE html>
<html>
<body>

  <nav>
    <a href="/html/">HTML</a> |
    <a href="/css/">CSS</a> |
    <a href="/js/">JavaScript</a> |
    <a href="/jquery/">jQuery</a>
  </nav>

</body>
</html>
```

## Output:

[HTML](#) | [CSS](#) | [JavaScript](#) | [jQuery](#)

# HTML <section> Element

- The <section> element defines a section in a document.
- Examples of where a <section> element can be used:
- Chapters
- Introduction
- News items
- Contact information

# Example

Two sections in a document:

## CODE:

```
<section>
```

```
<h1>WWF</h1>
```

```
<p>The World Wide Fund for Nature (WWF) is an international organization working on issues regarding the conservation, research and restoration of the environment, formerly named the World Wildlife Fund. WWF was founded in 1961.</p>
```

```
</section>
```

```
<section>
```

```
<h1>WWF's Panda symbol</h1>
```

```
<p>The Panda has become the symbol of WWF. The well-known panda logo of WWF originated from a panda named Chi Chi that was transferred from the Beijing Zoo to the London Zoo in the same year of the establishment of WWF.</p>
```

```
</section>
```

## Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Section Example</title>
</head>
<body>

  <section>
    <h1>WWF</h1>
    <p>
      The World Wide Fund for Nature (WWF) is an international
      organization working on issues regarding the conservation,
      research and restoration of the environment, formerly named
      the World Wildlife Fund. WWF was founded in 1961.
    </p>
  </section>

  <section>
    <h1>WWF's Panda symbol</h1>
    <p>
      The Panda has become the symbol of WWF. The well-known panda
      logo of WWF originated from a panda named Chi Chi that was
      transferred from the Beijing Zoo to the London Zoo in the same
      year of the establishment of WWF.
    </p>
  </section>

</body>
</html>
```

## Output:

### WWF

The World Wide Fund for Nature (WWF) is an international organization working on issues regarding the conservation, research and restoration of the environment, formerly named the World Wildlife Fund. WWF was founded in 1961.

### WWF's Panda symbol

The Panda has become the symbol of WWF. The well-known panda logo of WWF originated from a panda named Chi Chi that was transferred from the Beijing Zoo to the London Zoo in the same year of the establishment of WWF.

# HTML <article> Element

- The <article> element defines **independent, self-contained content**.
- The content inside an <article> should **make sense on its own**.
- An <article> can be **shared or distributed independently** from the rest of the webpage.
- It is commonly used for content that can stand alone.

## Examples of <article> content:

- Forum posts
- Blog posts
- User comments
- Product cards
- Newspaper or magazine articles
- News stories

```
<!DOCTYPE html>
<html>
<head>
  <title>Article Example</title>
</head>
<body>

  <h1>The article element</h1>

  <article>
    <h2>Google Chrome</h2>
    <p>
      Google Chrome is a web browser developed by Google, released in
      2008. Chrome is the world's most popular web browser today!
    </p>
  </article>

  <article>
    <h2>Mozilla Firefox</h2>
    <p>
      Mozilla Firefox is an open-source web browser developed by Mozilla.
      Firefox has been the second most popular web browser since January,
2018.
    </p>
  </article>

  <article>
    <h2>Microsoft Edge</h2>
    <p>
      Microsoft Edge is a web browser developed by Microsoft, released in
      2015. Microsoft Edge replaced Internet Explorer.
    </p>
  </article>

</body>
</html>
```

## Output:

### The article element

#### Google Chrome

Google Chrome is a web browser developed by Google, released in 2008. Chrome is the world's most popular web browser today!

#### Mozilla Firefox

Mozilla Firefox is an open-source web browser developed by Mozilla. Firefox has been the second most popular web browser since January, 2018.

#### Microsoft Edge

Microsoft Edge is a web browser developed by Microsoft, released in 2015. Microsoft Edge replaced Internet Explorer.

# HTML <aside> Element

- The <aside> element defines some content aside from the content it is placed in (like a sidebar).
- The <aside> content should be indirectly related to the surrounding content.

## Example

```
<!DOCTYPE html><html>
```

```
<body>
```

```
<p>
```

My family and I visited The Epcot center this summer.

The weather was nice, and Epcot was amazing!

I had a great summer together with my family!

```
</p>
```

```
<aside>
```

```
<h4>Epcot Center</h4>
```

```
<p>
```

Epcot is a theme park at Walt Disney World Resort featuring exciting attractions, international pavilions, award-winning fireworks and seasonal special events.

```
</p>
```

```
</aside>
```

```
</body>
```

```
</html>
```

## Output:

My family and I visited The Epcot center this summer. The weather was nice, and Epcot was amazing! I had a great summer together with my family!

### Epcot Center

Epcot is a theme park at Walt Disney World Resort featuring exciting attractions, international pavilions, award-winning fireworks and seasonal special events.

# HTML <footer> Element

The <footer> element defines a footer for a document or section.

A <footer> element typically contains:

- authorship information
- copyright information
- contact information
- sitemap
- back to top links
- related documents

# Example

## Code

```
<!DOCTYPE html>
<html>
<body>

    <footer>
        <p>
            <a href="mailto:cu@chitkara.edu.in">
                cu@chitkara.edu.in
            </a>
        </p>
    </footer>

</body>
</html>
```

## Output

[cu@chitkara.edu.in](mailto:cu@chitkara.edu.in)

# HTML `<figure>` and `<figcaption>` Elements

- The `<figure>` tag specifies self-contained content, like illustrations, diagrams, photos, code listings, etc.
- The `<figcaption>` tag defines a caption for a `<figure>` element. The `<figcaption>` element can be placed as the first or as the last child of a `<figure>` element.
- The `<img>` element defines the actual image/illustration.

# Example



## Code:

```
<!DOCTYPE html>
<html>
<body>

  <h2>Places to Visit</h2>

  <p>
    Puglia's most famous sight is the unique conical
    houses (Trulli)
    found in the area around Alberobello, a declared
    UNESCO World
    Heritage Site.
  </p>

  <figure>
    
    <figcaption>Fig.1 - Trulli, Puglia,
    Italy.</figcaption>
  </figure>

</body>
</html>
```

## Output:

### Places to Visit

Puglia's most famous sight is the unique conical houses (Trulli) found in the area around Alberobello, a declared UNESCO World Heritage Site.



Fig.1 - Trulli, Puglia, Italy.

# More Semantic Elements

|                                           |                                                                                             |
|-------------------------------------------|---------------------------------------------------------------------------------------------|
| <a href="#"><u>&lt;article&gt;</u></a>    | Defines independent, self-contained content                                                 |
| <a href="#"><u>&lt;aside&gt;</u></a>      | Defines content aside from the page content                                                 |
| <a href="#"><u>&lt;details&gt;</u></a>    | Defines additional details that the user can view or hide                                   |
| <a href="#"><u>&lt;figcaption&gt;</u></a> | Defines a caption for a <figure> element                                                    |
| <a href="#"><u>&lt;figure&gt;</u></a>     | Specifies self-contained content, like illustrations, diagrams, photos, code listings, etc. |
| <a href="#"><u>&lt;footer&gt;</u></a>     | Defines a footer for a document or section                                                  |
| <a href="#"><u>&lt;header&gt;</u></a>     | Specifies a header for a document or section                                                |
| <a href="#"><u>&lt;main&gt;</u></a>       | Specifies the main content of a document                                                    |
| <a href="#"><u>&lt;mark&gt;</u></a>       | Defines marked/highlighted text                                                             |
| <a href="#"><u>&lt;nav&gt;</u></a>        | Defines navigation links                                                                    |
| <a href="#"><u>&lt;section&gt;</u></a>    | Defines a section in a document                                                             |
| <a href="#"><u>&lt;summary&gt;</u></a>    | Defines a visible heading for a <details> element                                           |
| <a href="#"><u>&lt;time&gt;</u></a>       | Defines a date/time                                                                         |

# What are Non Semantic Elements?

Unlike, semantic elements they don't have any meaning. They don't tell anything about the content they contain. They can be used with different attributes to mark up semantics common to a group.

Following is the list of some non-semantic elements:

- div
- span

# Example

Example shows the use of non-semantic elements.

## Code:

```
<!DOCTYPE html>
<html>

<head>
  <title>my web page</title>
  <style type="text/css">
    span {
      color: green;
      font-size: 40px;
      font-weight: bold;
    }
  </style>
</head>

<body>
  <div>
    <span>GeeksForGeeks</span> <br>
    A computer science portal for geeks
  </div>
</body>

</html>
```

## Output:



HTML links are hyperlinks.

**Note:** A link does not have to be text. A link can be an image or any other HTML element!

**Syntax:**

```
<a href="url">link text</a>
```

- The most important attribute of the `<a>` element is the `href` attribute, which indicates the link's destination.
- The *link text* is the part that will be visible to the reader.
- Clicking on the link text, will send the reader to the specified URL address.

- The target attribute specifies where to open the linked document.
- The target attribute can have one of the following values:
- `_self` - Default. Opens the document in the same window/tab as it was clicked
- `_blank` - Opens the document in a new window or tab
- `_parent` - Opens the document in the parent frame
- `_top` - Opens the document in the full body of the window

## Example:

```
<a href="https://www.w3schools.com/" target="_blank">Visit  
W3Schools!</a>
```

# Absolute URLs vs. Relative URLs

## Example:

<h2>Absolute URLs</h2>

<p><a href="https://www.w3.org/">W3C</a></p>

<p><a href="https://www.google.com/">Google</a></p>

<h2>Relative URLs</h2>

<p><a href="html\_images.asp">HTML Images</a></p>

<p><a href="/css/default.asp">CSS Tutorial</a></p>

- To use an HTML button as a link, you have to add some JavaScript code.
- JavaScript allows you to specify what happens at certain events, such as a click of a button:

- **Example:**

```
<button onclick="document.location='default.asp'">HTML  
Tutorial</button>
```

# Tags in HTML – Image tag

## IMAGES

### <img> tag

The HTML `<img>` tag is used to **embed an image in a web page**.

- Closing tags are not used in HTML image element.

**The <img> tag has two required attributes:**

**src** - Specifies the path to the image

**alt** - Specifies an alternate text for the image



### Insert Image :

we can insert any image in web page by using `<img>` tag.

#### Syntax:

```
<img src = "Image URL" attributes-list>
```

#### Ex.:

```

```

# Tags in HTML – Image tag(Size)



## IMAGES:

### Set Image Width/Height

```
<img src = "/html/images/test.png" alt = "Test Image" width =  
"150" height = "100"/>
```



```
<html>  
<body>
```

```
<h2>Image Size</h2>
```

```

```

```
</body>  
</html>
```

# Tags in HTML – Image tag (Alignment)

## IMAGES

### IMAGE ALIGNMENT

- The align attribute specifies the **horizontal and vertical alignment** of the image in relation to the surrounding text/elements.

| Value  | Description                   |
|--------|-------------------------------|
| left   | Align the image to the left   |
| right  | Align the image to the right  |
| top    | Align the image on the top    |
| bottom | Align the image on the bottom |
| Middle | Align the image on the middle |

### Example

```
<img src = "test.png" alt = "Test Image" align = "right"/>
```

# Tags in HTML – Image tag (Borders)

## IMAGES:

### SET IMAGE BORDER:

- The `<img>` border attribute is used to specify the border width around the image.
- The default value of `<img>` border attribute is 0.
- It is not supported by HTML 5. Use CSS instead of this attribute

```
<img src = "test.png" alt = "Test Image" border = "3"/>
```

# The src Attribute

- The **src** (source) attribute specifies the **path (URL)** of the image.
- It is a **required attribute** of the **<img>** tag.
- The browser uses the src attribute to **locate and display the image**.
- The image file should remain in the **correct location** relative to the HTML file.
- If the image path is incorrect or the image is missing, the browser displays a **broken image icon**.
- When the image cannot be loaded, the **alt** text is displayed instead.
- The src attribute can contain:
  - **Relative URL** (e.g., images/photo.jpg)
  - **Absolute URL** (e.g., https://example.com/photo.jpg)

## Syntax:

```

```

- The required alt attribute provides an alternate text for an image, if the user for some reason cannot view it (because of slow connection, an error in the src attribute, or if the user uses a screen reader).
- The value of the alt attribute should describe the image:

**Syntax:**

```

```

# Image as a Link

To use an image as a link, put the `<img>` tag inside the `<a>` tag:

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<h2>Image as a Link</h2>
```

```
<p>The image is a link. You can click on it.</p>
```

```
<a href="default.asp">
```

```

```

```
</a>
```

```
</body>
```

```
</html>
```

**Output:**

## Image as a Link

The image is a link. You can click on it.



# Common Image Formats

| Abbreviation | File Format                           | File Extension                   |
|--------------|---------------------------------------|----------------------------------|
| APNG         | Animated Portable Network Graphics    | .apng                            |
| GIF          | Graphics Interchange Format           | .gif                             |
| ICO          | Microsoft Icon                        | .ico, .cur                       |
| JPEG         | Joint Photographic Expert Group image | .jpg, .jpeg, .jfif, .pjpeg, .jpp |
| PNG          | Portable Network Graphics             | .png                             |
| SVG          | Scalable Vector Graphics              | .svg                             |

- HTML tables allow web developers to arrange data into rows and columns.

| Company                      | Contact          | Country |
|------------------------------|------------------|---------|
| Alfreds Futterkiste          | Maria Anders     | Germany |
| Centro comercial Moctezuma   | Francisco Chang  | Mexico  |
| Ernst Handel                 | Roland Mendel    | Austria |
| Island Trading               | Helen Bennett    | UK      |
| Laughing Bacchus Winecellars | Yoshi Tannamuri  | Canada  |
| Magazzini Alimentari Riuniti | Giovanni Rovelli | Italy   |

# HTML `<table>` Tag

An HTML table consists of one `<table>` element and one or more `<tr>`, `<th>`, `<td>` elements.

The `<tr>` element defines a table row,  
the `<th>` element defines a table header,  
and the `<td>` element defines a table cell.

# Example

```
<h1>The table element</h1>
<table style = “width:500px”>
<table style = “float:right”>
  <tr>
    <th>Month</th>
    <th>Savings</th>
  </tr>
  <tr>
    <td>January</td>
    <td>$100</td>
  </tr>
  <tr>
    <td>February</td>
    <td>$80</td>
  </tr>
</table>
</body>
</html>
```

## Output:

| Month   | Savings |
|---------|---------|
| January | \$100   |
| Feb     | \$80    |

# HTML <caption> Tag

The <caption> tag defines a table caption.

The <caption> tag must be inserted immediately after the <table> tag.

```
<!DOCTYPE html>
<html>
<head>
<style>
table, th, td {
    border: 1px solid black;
}
body
{
background-image: url('image.jpg');
}
</style>
</head>
<body>
<table>
  <caption style="text-align:right">My savings</caption>
  <tr>
    <th>Month</th>
    <th>Savings</th>
  </tr>
  <tr>
    <td>January</td>
    <td>$100</td>
  </tr>
</table>
<br>
```

| My savings |         |
|------------|---------|
| Month      | Savings |
| January    | \$100   |

- The <colgroup> tag specifies a group of one or more columns in a table for formatting.
- The <colgroup> tag is useful for applying styles to entire columns, instead of repeating the styles for each cell, for each row.
- Note: The <colgroup> tag must be a child of a <table> element, after any <caption> elements and before any <thead>, <tbody>, <tfoot>, and <tr> elements.

# How To Add a Border

- To add a border, use the CSS border property on table, th, and td elements:
- `table, th, td {  
 border: 1px solid black;  
}`

## Collapse Borders

- `table, th, td {  
 border: 1px solid black;  
 border-collapse: collapse;  
}`

**Cell padding is the space between the cell edges and the cell content.**

- By default the padding is set to 0.
- To add padding on table cells, use the CSS padding property:

- Example

```
th, td {  
padding: 15px;  
}
```

- Cell spacing **is the space between each cell.**
- **By default the space is set to 2 pixels.**
- To change the space between table cells, use the CSS border-spacing property on the table element:
- Example
- `table {`
- `border-spacing: 30px;`
- `}`

# Tables – bordercolor



- we can also set border color also using **bordercolor** attribute.

| Name   | Salary    |
|--------|-----------|
| Shivam | 50,000.00 |
| Vansh  | 45,000.00 |

```
<html>
<head>
<title>HTML Tables</title> </head>
<body>
<table border="1" bordercolor="red" >
<tr>
    <th>Name</th>
    <th>Salary</th>
</tr>
<tr>
    <td>Shivam</td>
    <td>50,000.00</td>
</tr>
<tr>
    <td>Vansh</td>
    <td>45,000.00</td>
</tr>
</table>
</body>
</html>
```

## Tables Backgrounds

- **bgcolor attribute** – we can set background color for whole table or just for one cell.
- **background attribute** – we can set background image for whole table or just for one cell.

| Name    | Salary    |
|---------|-----------|
| Jayapal | 50,000.00 |
| Ravi    | 45,000.00 |

```
<html>
<head>
<title>HTML Tables</title> </head>
<body>
<table border="1" bgcolor="yellow">
<tr>
<th>Name</th>
<th>Salary</th>
</tr>
<tr>
<td>Jayapal</td>
<td>50,000.00</td>
</tr>
<tr>
<td>Ravi</td>
<td>45,000.00</td>
</tr>
</table>
</body>
</html>
```

# Tables – table body <tbody>



- HTML **<tbody>** tag is used to group the table rows (<tr>) together, which indicates that this is body part of a table (<table>).
- The **<tbody>** tag must be a child of <table> element.
- The **<tbody>** is used along with **<thead>** and **<tfoot>** which shows the different part of the table that are table head, table body, and table footer, however, it does not affect the layout of the table.

| Day | venue |
|-----|-------|
| Mon | TG    |
| Tue | PG    |
| wed | tesla |

```
<!DOCTYPE html>
<html>
<h1>The thead, tbody, and tfoot elements</h1>
<table border="1">
  <thead>
    <tr>
      <th>Day</th>
      <th>venue</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>Mon</td>
      <td>TG</td>
    </tr>
    <tr>
      <td>Tue</td>
      <td>PG</td>
    </tr>
  </tbody>
```

```
    <tfoot>
      <tr>
        <td>wed</td>
        <td>tesla</td>
      </tr>
    </tfoot>
  </table>

</body>
</html>
```

# Tables- Height and Width attributes



- We can set a table **width** and **height** using **width** and **height** attributes.
- Specify table width or height in terms of **pixels** or in terms of **percentage** of available screen area.

## Table Caption

- The caption tag will serve as a title or explanation for the table and it shows up at the top of the table. This tag is deprecated in newer version of HTML/XHTML

This is the caption

|                 |                 |
|-----------------|-----------------|
| Row 1, Column 1 | Row 1, Column 2 |
| Row 2, Column 1 | Row 2, Column 2 |

```
<html>

<head>
  <title>HTML Table Width/Height</title>
</head>

  <body>
    <table border = "1" width = "400" height =
    "150">
      <caption>This is the caption</caption>
      <tr>
        <td>Row 1, Column 1</td>
        <td>Row 1, Column 2</td>
      </tr>

      <tr>
        <td>Row 2, Column 1</td>
        <td>Row 2, Column 2</td>
      </tr>
    </table>
  </body>

</html>
```

# Define an HTML Table

## Code:

```
<table>
  <tr>
    <th>Company</th>
    <th>Contact</th>
    <th>Country</th>
  </tr>
  <tr>
    <td>Alfreds Futterkiste</td>
    <td>Maria Anders</td>
    <td>Germany</td>
  </tr>
  <tr>
    <td>Centro comercial Moctezuma</td>
    <td>Francisco Chang</td>
    <td>Mexico</td>
  </tr>
</table>
```

## Output:

| Company                    | Contact         | Country |
|----------------------------|-----------------|---------|
| Alfreds Futterkiste        | Maria Anders    | Germany |
| Centro comercial Moctezuma | Francisco Chang | Mexico  |

# Table Cells

Each table cell is defined by a `<td>` and a `</td>` tag.  
td stands for table data.

## Code:

```
<table>  
  <tr>  
    <td>Emil</td>  
    <td>Tobias</td>  
    <td>Linus</td>  
  </tr>  
</table>
```

## Output:

**TD elements define table cells**

|      |        |       |
|------|--------|-------|
| Emil | Tobias | Linus |
|------|--------|-------|

tr stands for table row.

## Code:

```
<table>
  <tr>
    <td>Emil</td>
    <td>Tobias</td>
    <td>Linus</td>
  </tr>
  <tr>
    <td>16</td>
    <td>14</td>
    <td>10</td>
  </tr>
</table>
```

## Output:

|      |        |       |
|------|--------|-------|
| Emil | Tobias | Linus |
| 16   | 14     | 10    |

# Table Headers

th stands for table header.

## Code:

```
<table>
  <tr>
    <th>Person 1</th>
    <th>Person 2</th>
    <th>Person 3</th>
  </tr>
  <tr>
    <td>Emil</td>
    <td>Tobias</td>
    <td>Linus</td>
  </tr>
  <tr>
    <td>16</td>
    <td>14</td>
    <td>10</td>
  </tr>
</table>
```

## Output:

| Person 1 | Person 2 | Person 3 |
|----------|----------|----------|
| Emil     | Tobias   | Linus    |
| 16       | 14       | 10       |

# Table Tags

| Tag                                     | Description                                                             |
|-----------------------------------------|-------------------------------------------------------------------------|
| <a href="#"><u>&lt;table&gt;</u></a>    | Defines a table                                                         |
| <a href="#"><u>&lt;th&gt;</u></a>       | Defines a header cell in a table                                        |
| <a href="#"><u>&lt;tr&gt;</u></a>       | Defines a row in a table                                                |
| <a href="#"><u>&lt;td&gt;</u></a>       | Defines a cell in a table                                               |
| <a href="#"><u>&lt;caption&gt;</u></a>  | Defines a table caption                                                 |
| <a href="#"><u>&lt;colgroup&gt;</u></a> | Specifies a group of one or more columns in a table for formatting      |
| <a href="#"><u>&lt;col&gt;</u></a>      | Specifies column properties for each column within a <colgroup> element |
| <a href="#"><u>&lt;thead&gt;</u></a>    | Groups the header content in a table                                    |
| <a href="#"><u>&lt;tbody&gt;</u></a>    | Groups the body content in a table                                      |
| <a href="#"><u>&lt;tfoot&gt;</u></a>    | Groups the footer content in a table                                    |

# Tags in HTML – Lists (Definition lists)

## DEFINITION LISTS

### Definition Lists:

- HTML supports a **list style** which is called definition lists where entries are listed like in a **dictionary or encyclopedia**.
- The definition list is the ideal way to present a glossary, list of terms, or other name/value list.

Definition List makes use of following three tags.

- 1). **<dl>** - Defines the start of the list  
**</dl>** - Defines the end of the list
- 2). **<dt>** - A term
- 3). **<dd>** - Term definition

#### HTML

This stands for Hyper Text Markup Language

#### HTTP

This stands for Hyper Text Transfer Protocol

```
<html>
  <head>
    <title>HTML Definition List</title>
  </head>
  <body>
    <dl>
      <dt>HTML</dt>
      <dd>This stands for Hyper Text Markup
      Language</dd>
      <dt>HTTP</dt>
      <dd>This stands for Hyper Text Transfer
      Protocol</dd>
    </dl>
  </body>
</html>
```

## LISTS

- A list is a **collection of items**.
- HTML offers web authors three ways for specifying lists of information.
- All lists must contain one or more list elements

### Lists are of three types

**1. Un ordered list** `<ul>` Items **marked with bullets**.

- html
- css
- javascript

**2. Ordered List** `<ol>` Items marked with **numbers or letters**

- 1.Coffee
- 2.Tea
- 3.Milk

**3. Definition list** `<dl>` Arranges items in the same way as they are arranged in a **dictionary**.

- Coffee  
- black hot drink
- Milk  
- white cold drink

# Tags in HTML – Lists (Unordered lists)

## LISTS

### Unordered Lists :

- An unordered list is a **collection of related items that have no special order or sequence.**
- This list is created by using HTML **<ul>** tag.
- Each item in the list is marked with a **bullet.**

#### Example

```
<html>
  <head>
    <title>HTML Unordered List</title>
  </head>
  <body>
    <ul> <li>Beetroot</li>
        <li>Ginger</li>
        <li>Potato</li>
        <li>Radish</li>
    </ul>
  </body>
</html>
```

- Beetroot
- Ginger
- Potato
- Radish

# Tags in HTML – Lists (Unordered lists)

## UNORDERD LISTS WITH “TYPE” ATTRIBUTE

### The type Attribute

we can use type attribute for <ul> tag to specify the **type** of bullet we like. By default, it is a **disc**.

Following are the possible options –

**<ul type = "square"> <ul type = "disc"> <ul type = "circle">**

### Example

```
<html>
<head> <title>HTML Unordered List</title>
</head>
<body>
  <ul type = "square">
    <li>Beetroot</li>
    <li>Ginger</li>
    <li>Potato</li>
    <li>Radish</li>
  </ul>
</body>
</html>
```

- Beetroot
- Ginger
- Potato
- Radish

# Tags in HTML – Lists (Ordered lists)

## ORDERED LISTS

### Ordered Lists:-

- Items are **numbered** list instead of bulleted, This list is created by using **<ol>** tag.
- The **numbering starts at one** and is incremented by one for each successive ordered list element tagged with <li>.

#### Example

```
<!DOCTYPE html>
<html>
  <head>
    <title>HTML Ordered List</title>
  </head>
  <body>
    <ol>
      <li>Beetroot</li>
      <li>Ginger</li>
      <li>Potato</li>
      <li>Radish</li>
    </ol>
  </body>
</html>
```

1. Beetroot
2. Ginger
3. Potato
4. Radish

# Tags in HTML – Lists (Ordered lists)

## ORDERED LISTS WITH TYPE ATTRIBUTE

### The type Attribute:

- we can use **type** attribute for `<ol>` tag to specify the type of numbering we like. By default, it is a number. Following are the possible options.

`<ol type = "1">` - Default-Case Numerals.

`<ol type = "I">` - Upper-Case Numerals.

`<ol type = "i">` - Lower-Case Numerals.

`<ol type = "A">` - Upper-Case Letters.

`<ol type = "a">` - Lower-Case Letters.

```
<html>
<head> <title>HTML Ordered List</title> </head>
<body>
  <ol type = "1">
    <li>Beetroot</li>
    <li>Ginger</li>
    <li>Potato</li>
    <li>Radish</li>
  </ol>
</body>
</html>
```

1. Beetroot
2. Ginger
3. Potato
4. Radish

# Tags in HTML – Lists (Ordered lists)

## ORDERED LISTS WITH “START” ATTRIBUTE

### The start Attribute

we can use **start** attribute for <ol> tag to specify the starting point of numbering we need. Following are the possible options –

**<ol type = "1" start = "4"> - Numerals starts with 4.**

**<ol type = "I" start = "4"> - Numerals starts with IV.**

**<ol type = "i" start = "4"> - Numerals starts with iv.**

**<ol type = "a" start = "4"> - Letters starts with d.**

**<ol type = "A" start = "4"> - Letters starts with D.**

```
<html>
<head> <title>HTML Ordered List</title> </head>
<body>
<ol type = "i" start = "4">
  <li>Beetroot</li>
  <li>Ginger</li>
  <li>Potato</li>
  <li>Radish</li>
</ol>
</body>
</html>
```

```
iv. Beetroot
v. Ginger
vi. Potato
vii. Radish
```



Multimedia tags include:

- Image
- Video
- Audio
- Flash Files
- Marquee

**1.Image Tag:** The `<img>` [HTML](#) element embeds an image into the document. It has various attributes such as **src** attribute is required and contains path to the image you want to embed.

**alt** (alternative text) attribute holds a textual replacement for the image, which is mandatory and incredibly useful for accessibility — screen readers read the attribute value out to their users so they know what the image means.

Alt text is also displayed on the page if the image can't be loaded for some reason: for example, network errors, content blocking etc.

**hspace: (Horizontal space)** The number of pixels of white space on the left and right of the image.

Use the [margin](#) CSS property instead.

**vspace: (vertical space)** The number of pixels of white space above and below the image.

Use the [margin](#) CSS property instead.

**width/height:** The intrinsic width/height of the image in pixels. Must be an integer without a unit.

```

```

### USE IMAGE DIRECTLY FROM INTERNET WITHOUT DOWNLOADING

Right click on image and click *copy image address* and paste in **src** attribute

## 2. Video

```
<video width="320" height="240" controls autoplay muted loop>  
    <source src="Colourmixing.mp4" type="video/mp4">  
    <source src="movie.ogg" type="video/ogg">
```

Your browser does not support the video tag.  
</video>

The **<video>** tag is used to embed video content in a document. The `<video>` tag contains one or more `<source>` tags with different video sources. The browser will choose the first source it supports. The text between the `<video>` and `</video>` tags will only be displayed in browsers that do not support the `<video>` element. There are three supported video formats in HTML: MP4, WebM, and OGG.

### Attributes:

**Controls:** Specifies that video controls should be displayed (such as a play/pause button etc).

**Loop:** Specifies that the video will start over again & again, every time it is finished. We cannot set the number of times a video is played i.e. No control over loop attribute but we can use JavaScript to set loops.

**Autoplay:** Specifies that the video will start playing upon page load. Many browsers can only autoplay the videos with sound off, so you'll need to add *muted* attribute

**Muted:** specifies that the audio output of the video should be muted. Controls, loop, autoplay and loop are boolean attributes.

### Code to include audio file:

```
<audio controls loop>  
  <source src="abc.m4a">  
  <source src="horse.mp3" type="audio/mpeg">  
  Your browser does not support the audio tag.  
</audio>
```

### Code to include flash/animated file:

```
<embed src="ball.svg" width="500" height="450">  
Svg: Scalable vector Graphics
```

## 5. Marquee

It is used to scroll an image or text horizontally or vertically. But this tag has been **deprecated in HTML 5**.

```
<marquee> WELCOME </marquee>
```

```
<marquee bgcolor="lightgreen" font size="5"  
direction="right" >  
WELCOME </marquee>
```

# Attributes of Marquee Tag

| Attribute    | Values                   | Description                                                                                                                                                                                                                   |
|--------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direction    | left, right, up, down    | Specifies the direction in which the marquee content scrolls.                                                                                                                                                                 |
| Behavior     | scroll, slide, alternate | Defines the type of movement of the marquee. 'scroll' is the default behavior, causing the content to scroll endlessly. 'slide' makes the content slide in and stop. 'alternate' causes the content to bounce back and forth. |
| scrollamount | number                   | Defines the speed of the marquee text. Higher values result in faster scrolling. Default=6                                                                                                                                    |
| Scrolldelay  | number                   | Sets the delay in milliseconds between each movement. Higher values result in slower scrolling.                                                                                                                               |
| Loop         | number                   | Specifies the number of times the marquee should loop. The default is infinite.                                                                                                                                               |
| Bgcolor      | color                    | Sets the background color of the marquee. You can use color names or hex codes.                                                                                                                                               |
| Width        | value (in px, %, etc.)   | Sets the width of the marquee. It can be in pixels, percentages, etc.                                                                                                                                                         |
| Height       | value (in px, %, etc.)   | Sets the height of the marquee. It can be in pixels, percentages, etc.                                                                                                                                                        |

## Behavior: It has 3 values:

- **Scroll:** It is default property. It is used to scroll the text from **right to left**, and restarts at the right side of the marquee when it is reached to the end of left side.
- **Slide:** In this marquee, all the contents to be scrolled will slide the entire length of marquee but stops at the end to display the content permanently.
- **Alternate:** It scrolls the text from **right to left** and goes back **left to right**.

```
<marquee bgcolor="lightgreen"
        behavior="slide"      direction="right"
        scrollamount="25"     width="60%"
        height="200"
        hspace="250" vspace="200"    loop="4">
```

# FORMS



## FORMS:

An HTML form is a **section of a document** which **contains controls** such as

- **text fields,**
- **password fields,**
- **checkboxes,**
- **radio buttons,**
- **submit button,**
- **menus etc.**

An example of an HTML form with a red border. It contains the following elements: a 'Name' label followed by a text input field with placeholder text 'Name'; a 'Password' label followed by a password input field with placeholder text 'Password'; a 'Gender' label followed by two radio buttons labeled 'Male' and 'Female'; an 'Email' label followed by a text input field with placeholder text 'Email'; a 'Phone no' label followed by a dropdown menu showing '977' and a text input field containing '98480\*\*\*\*\*'; and a 'Submit' button at the bottom left.

- An HTML form facilitates the **user to enter data** that is to be sent to the server for processing such as name, email address, password, phone number, etc. .
- There are various **form elements** available like text fields, text area fields, drop-down menus, radio buttons, checkboxes, etc.

## HTML Form Tags:

### <form>

The HTML <form> element **defines a form** that is used to collect user input:

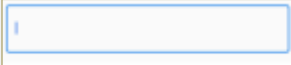
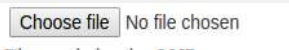
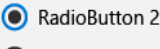
```
<form action="server url" method="get|post">
    //input controls e.g. textfield, textarea, radiobutton, button
</form>
```

| Sr.N<br>0 | Attribute & Description                                                                                                                               |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1         | <b>action</b> : Backend script ready to process your passed data.                                                                                     |
| 2         | <b>method</b> : It to be used to upload data. The most frequently used are GET and POST methods.                                                      |
| 3         | <b>target</b> : Specify the target window or frame where the result of the script will be displayed. It takes values like _blank, _self, _parent etc. |

## FORMS:

**<input>** - It defines an **input control**.

The **<input>** element can be displayed in several ways, depending on the **type** attribute.

| Type                                 |                                                                                     | Description                                                                            |
|--------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| <b>&lt;input type="text"&gt;</b>     |    | Defines a one-line <b>text input field</b>                                             |
| <b>&lt;input type="password"&gt;</b> |    | Defines a <b>password field</b>                                                        |
| <b>&lt;input type="checkbox"&gt;</b> |    | Defines a <b>checkbox</b>                                                              |
| <b>&lt;input type="button"&gt;</b>   |    | Defines a <b>clickable button</b> (mostly used with a JavaScript to activate a script) |
| <b>&lt;input type="file"&gt;</b>     |    | Defines a <b>file-select field</b> and a "Browse" button (for file uploads)            |
| <b>&lt;input type="radio"&gt;</b>    |  | Defines a <b>radio button</b> (for selecting one of many choices)                      |
| <b>&lt;input type="submit"&gt;</b>   |  | Defines a <b>submit button</b> (for submitting the form)                               |

---

## FORMS:

---

### HTML Form Controls

There are different types of form controls that you can use to collect data using HTML form –

1. Text field
  2. Password Field
  3. Text Area
  4. Checkbox
  5. Radio Box
  6. Select Box
  7. File Select boxes
  8. Buttons
-

## FORMS:

### 1. Text field (or) Single-line text input control

This control is used for items that require only *one line of user input*, such as search boxes or names. They are created using HTML **<input>** tag.

| Sr.No            | Attribute & Description                                                                       |
|------------------|-----------------------------------------------------------------------------------------------|
| <b>type</b>      | Indicates the type of <b>input control</b> and for text input.                                |
| <b>name</b>      | Used to give a <b>name</b> to the control                                                     |
| <b>value</b>     | This can be used to provide an <b>initial value</b>                                           |
| <b>size</b>      | Allows to specify the <b>width of the text-input</b> control in terms of characters.          |
| <b>maxlength</b> | Allows to specify the <b>maximum number of characters</b> a user can enter into the text box. |

```
<html>
<head>
  <title>Text Input Control</title>
</head>
<body>
  <form >
    First name: <input type = "text" name = "first_name" />
    <br>
    Last name: <input type = "text" name = "last_name" />
  </form>
</body>
</html>
```

sample.html x Inbox (4) - j... x

file:///E:/jayapal/r

First name:

Last name:

---

## FORMS:

---

### 2. Password Field

- This is also a **single-line text** input but it masks the character as soon as a user enters it.
- They are also created using HTML `<input>` tag but type attribute is set to **password**.

User ID :

Password:

```
<html>
<head> <title>Password Input Control</title> </head>
<body>
<form >
  User ID : <input type = "text" name = "user_id" />
  <br> Password:
  <input type = "password" name = "password" />
</form>
</body>
</html>
```



---

## FORMS:

---

### 3. Text Area (or) Multiple-Line Text Input Control

This is used when the user is required to give details that may be longer than a single sentence. Multi-line input controls are created using HTML **<textarea>** tag.

Description :

```
<html>
<head>
  <title>Multiple-Line Input Control</title>
</head>

<body>
  <form>
    Description :<br>
    <textarea rows = "5" cols = "50" name = "description">
      Enter description here...
    </textarea>
  </form>
</body>

</html>
```

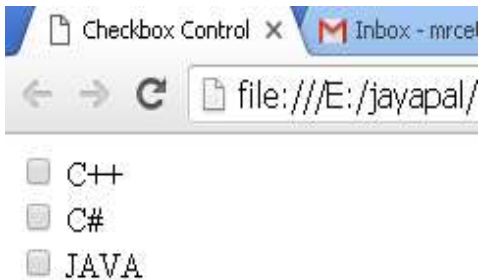
---

**FORMS:**

---

**4. Checkbox Control**

Checkboxes are used when more than one option is required to be selected. They are also created using HTML `<input>` tag but type attribute is set to checkbox.



```
<html> <head> <title>Checkbox Control</title> </head>
<body>
<form>
<input type="checkbox" name="C++" value="on"> C++
<br>
<input type="checkbox" name="C#" value="on"> C#
<br>
<input type="checkbox" name="JAVA" value="on"> JAVA
</form>
</body> </html>
```

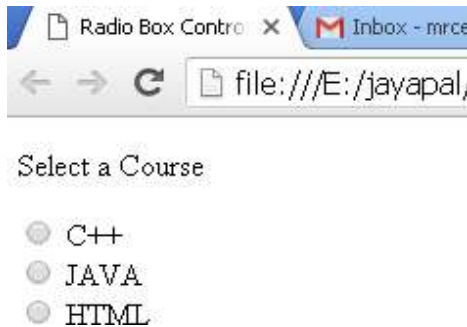
---

**FORMS:**

---

**5. Radio Button Control**

Radio buttons are used when out of many options, just one option is required to be selected. They are also created using HTML `<input>` tag but type attribute is set to radio



```
<html>
<head>
<title>Radio Box Control
</title> </head>
<body> <p>Select a Course</p>
<form>
<input type="radio" name="subject" value="C++"> C++
<br>
<input type="radio" name="subject" value="JAVA"> JA VA
<br>
<input type="radio" name="subject" value="HTML"> HTML
</form>      </body> </html>
```



---

## FORMS:

---

### 6. Select Box Controls

A **select box**, also called **drop down box** which provides option to list down various options in the form of drop down list, from where a user can select one or more options



Select Box Control



```
<html>
<head>
<title>Select Box Control</title>
</head>
<body>
<form>
<select name="dropdown">
<option value="C++" selected>C++</option>
<option value="JAVA">JA VA</option>
<option value="HTML">HTML</option>
</select>
</form>
</body>
</html>
```



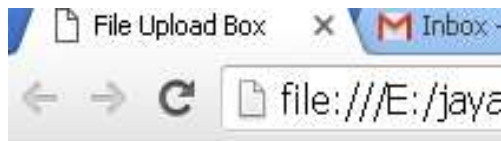
---

## FORMS:

---

### 7. File Select boxes

If we want to allow a user to **upload a file** to web site, we will need to use a file upload box, also known as a file select box. This is also created using the `<input>` element but type attribute is set to file.



File Upload Box

Choose File No file chosen

```
<html>
<head>
<title>File Upload Box</title>
</head>
<body>
<p>File Upload Box</p>
<form>
<input type="file" name="fileupload" >
</form>
</body>
</html>
```

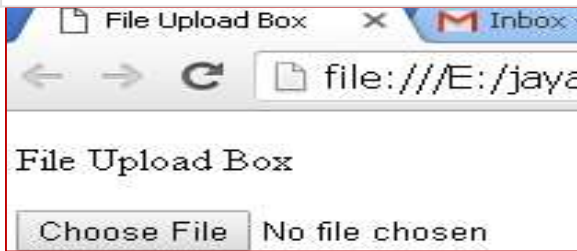
## FORMS

### 8. Buttons

There are various ways in HTML to create clickable buttons.

we can also create a clickable button using **<input>** tag by setting its type attribute to **button**.

| Sr.No         | Type & Description                                                                                   | ind values:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>submit</b> | This creates a button that automatically <b>submits</b> a form.                                      | <pre> &lt;html&gt; &lt;head&gt; &lt;title&gt;File Upload Box&lt;/title&gt; &lt;/head&gt; &lt;body&gt; &lt;form&gt; &lt;input type="<b>submit</b>" name="<b>submit</b>" value="<b>Submit</b>" /&gt; &lt;input type="<b>reset</b>" name="<b>reset</b>" value="<b>Reset</b>" /&gt; &lt;input type="<b>button</b>" name="<b>ok</b>" value="<b>OK</b>" /&gt; &lt;input type="<b>image</b>" name="<b>imagebutton</b>" src="<b>test1.png</b>" /&gt; &lt;/form&gt; &lt;/body&gt; &lt;/html&gt; </pre> |
| <b>reset</b>  | This <b>creates a button that automatically resets</b> form controls to their initial values.        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>button</b> | This creates a button that is used to trigger a client-side script when the user clicks that button. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>image</b>  | This creates a <b>clickable button but we can use an image</b> as background of the button.          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |



---

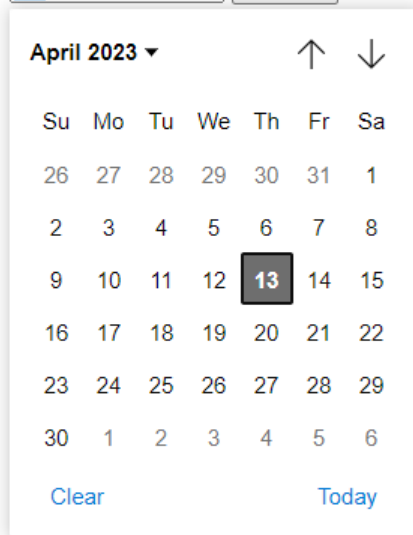
## Displaying date picker

---

### 9. HTML `<input type="date">`

- The `<input type="date">` defines a date picker.
- The resulting value includes the year, month, and day.attribute

DOB:



April 2023 ▾ ↑ ↓

| Su | Mo | Tu | We | Th | Fr | Sa |
|----|----|----|----|----|----|----|
| 26 | 27 | 28 | 29 | 30 | 31 | 1  |
| 2  | 3  | 4  | 5  | 6  | 7  | 8  |
| 9  | 10 | 11 | 12 | 13 | 14 | 15 |
| 16 | 17 | 18 | 19 | 20 | 21 | 22 |
| 23 | 24 | 25 | 26 | 27 | 28 | 29 |
| 30 | 1  | 2  | 3  | 4  | 5  | 6  |

Clear Today

```
<html>
<body>

<h1>Show a Date Control</h1>

<form>
  DOB: <input type="date" id="birthday" name="birthday">
  <input type="submit">
</form>

</body>
</html>
```

## Displaying date picker with dd/mm/yyyy format and ranges

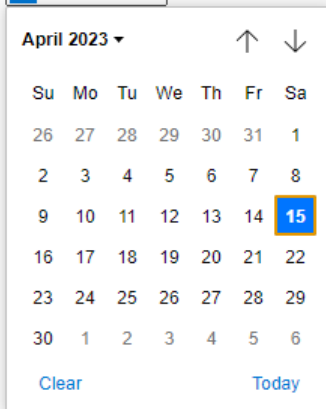
### 10. HTML <input type="date">

- To set and get the input type date in **dd-mm-yyyy** format we will use attribute **placeholder="dd-mm-yyyy"**.
- We can also mention the **start** and **end dates** for the date picker with **min** and **max** attributes.

#### Date format

Get input date in dd-mm-yyyy format

Enter the Date:



| April 2023 |    |    |    |    |    |    |
|------------|----|----|----|----|----|----|
| Su         | Mo | Tu | We | Th | Fr | Sa |
| 26         | 27 | 28 | 29 | 30 | 31 | 1  |
| 2          | 3  | 4  | 5  | 6  | 7  | 8  |
| 9          | 10 | 11 | 12 | 13 | 14 | 15 |
| 16         | 17 | 18 | 19 | 20 | 21 | 22 |
| 23         | 24 | 25 | 26 | 27 | 28 | 29 |
| 30         | 1  | 2  | 3  | 4  | 5  | 6  |

Clear Today

```
<html>
<body>
  <h1>Date format</h1>
  <h3> Get input date in
  dd-mm-yyyy format </h3>
  <label for="start"> Enter the Date: </label>
  <input type="date" name="start"
  placeholder="dd-mm-yyyy"
  min="1997-01-01" max="2030-12-31">
</body>
</html>
```

## Displaying time picker

### 11. HTML `<input type="time">`

- The `<input type="date">` defines a date picker.  
`placeholder="dd-mm-yyyy"`.
- We can also mention the **start** and **end dates** for the date picker with `min` and `max` attributes.

### Time Picker

Choose a time for your meeting:

```
<html>
<body>
<label for="appt">Choose a time for your meeting:</label>
<input type="time" id="appt" name="appt"
      min="09:00" max="18:00" required>
<small>Office hours are 9am to 6pm</small>
</body>
</html>
```



---

## Telephone number

---

### 12. `<input type="tel">`

The `<input type="tel">` defines a telephone number.

#### Time Picker

Enter your phone number:

Format: 91-456-7890

```
<html>
<body>
<label for="appt">Choose a time for your meeting:</label>
<input type="time" id="appt" name="appt"
      min="09:00" max="18:00" required>
<small>Office hours are 9am to 6pm</small>
</body>
</html>
```



## RANGE

`<input type="range" id="a" value="50">`



## The input required attribute

```
<!DOCTYPE html>
<html>
<body>
<h1>The input required attribute</h1>
<form action="/action_page.php">
  <label for="username">Username:</label>
  <input type="text" id="username" name="username" required>
  <input type="submit">
</form>
</body>
</html>
```

## The input required attribute

Username:

! Please fill in this field.

Lastname: `<input type="text" name="lastname" value="sharma" readonly>`

Lastname:

- `<input type="hidden" name="hiddenField" value="nyah">`  
&lt;-- right there, don't you see it?

A hidden field: <-- right there, don't you see it?

- What good is this?
  - All **input** fields are sent back to the server, including hidden fields
  - This is a way to include information that the user doesn't need to see (or that you don't want him/her to see)
  - The **value** of a hidden field can be set programmatically (by JavaScript) before the form is submitted

## HTML <fieldset> Tag

The <fieldset> tag is used to group related elements in a form.  
The <fieldset> tag draws a box around the related elements.

```
<!DOCTYPE html>
<html>
<body>
```

```
<h1>The fieldset element</h1>
```

```
<form action="/action_page.php">
  <fieldset>
    <legend>Personalia:</legend>
    <label for="fname">First name:</label>
    <input type="text" id="fname" name="fname"><br><br>
    <label for="lname">Last name:</label>
    <input type="text" id="lname" name="lname"><br><br>
    <label for="email">Email:</label>
    <input type="email" id="email" name="email"><br><br>
    <label for="birthday">Birthday:</label>
    <input type="date" id="birthday" name="birthday"><br><br>
    <input type="submit" value="Submit">
  </fieldset>
</form>
</body>
</html>
```

### The fieldset element

Personalia:
 

First name:

Last name:

Email:

Birthday:



## The fieldset element

Personalia:

First name:

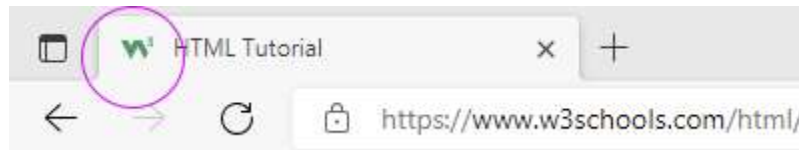
Last name:

Email:

Birthday:



- A favicon is a small image displayed next to the page title in the browser tab.
- **HTML Favicon** is a small icon representing a website, appearing in the browser's tab or bookmark bar. Defined using the `<link>` tag with ``rel="icon"``, it is also known as the Favorite icon.



# Adding a Favicon

## How to add favicon in HTML?

Add a `<link>` element to your "html" file, after the `<title>` element, like this:

```
<head>  
    <title> Adding Favicon </title>  
    <link rel="icon" type="image" href="shopping.png">  
</head>
```

`<link>` Defines the relationship between a document and an external resource

## Favicon

A favicon is a small image displayed next to the page title in the browser tab.

```
<!DOCTYPE html>
<html>
<head>
  <title>My Page Title</title>
  <link rel="icon" type="image" href="/images/favicon.ico">
</head>
<body>

<h1>This is a Heading</h1>
<p>This is a paragraph.</p>

</body>
</html>
```

# Meta Tag

- The `<meta>` tag defines metadata about an HTML document. Metadata is data (information) about data.
- `<meta>` tags always go inside the `<head>` element, and are typically used to specify character set, page description, keywords, author of the document, and viewport settings.
- Metadata will not be displayed on the page, but is machine parsable.
- Metadata is used by browsers (how to display content or reload page), search engines (keywords), and other web services.
- There is a method to let web designers take control over the viewport (the user's visible area of a web page), through the `<meta>` tag.

```
<head>  
<meta charset="UTF-8">  
<meta name="keywords" content="HTML, CSS, JavaScript">  
<meta name="description" content="Free Web tutorials for HTML  
and CSS">  
<meta name="author" content="John Doe">  
</head>
```

UTF-8: UTF-8 stands for “Unicode Transformation Format - 8 bits.” UTF-8 is an encoding system for Unicode. It can translate any Unicode character to a matching unique binary string, and can also translate the binary string back to a Unicode character.

- **SEO (Search Engine Optimization) in Points**
- **SEO** stands for **Search Engine Optimization**.
- SEO helps **search engines understand** your website content.
- It also helps **users find your website** through search engines like Google.
- A well-optimized website has a **better chance of appearing in search results**.
- SEO improves a website's **visibility and ranking** in search engines.
- Following **Google Search Essentials** increases the chances of your website being indexed.
- There is **no guarantee** that a website will rank first on Google.
- SEO involves improving a website so that search engines can:
  - Crawl the website
  - Index the pages
  - Understand the content
- Good SEO practices make your website **more accessible** to search engines and users.
- SEO is an **ongoing process** that requires regular improvements and updates.
- Best SEO practices benefit **all search engines**, not just Google.

## Scenario

- A student creates a website about Java programming.

```
<!DOCTYPE html>
<html>
<head>
  <title>Page</title>
</head>
<body>
  <h1>Java</h1>
  <p>Learn Java programming.</p>
</body>
</html>
```

## Problems

- Generic title ("Page")
- No meta description
- No keywords
- No viewport tag
- Difficult for search engines to understand the page

## Optimized Website

```
<meta charset="UTF-8">

<meta name="description"
      content="Free Java programming tutorials for beginners.">

<meta name="keywords"
      content="Java, Programming, HTML, Coding">

<meta name="author"
      content="Money">

<meta name="viewport"
      content="width=device-width, initial-scale=1.0">
</head>

<body>
  <h1>Learn Java Programming</h1>

  <p>Start learning Java from basic to advanced concepts.</p>
</body>
</html>
```

# Before VS After

| <b>Before SEO</b>   | <b>After SEO</b>                      |
|---------------------|---------------------------------------|
| Generic title       | Descriptive title                     |
| No meta description | Meta description added                |
| No keywords         | Keywords included                     |
| No viewport         | Mobile-friendly                       |
| Hard to understand  | Easy for search engines to understand |
| Lower visibility    | Better search visibility              |

- Increases website visibility
  - Attracts more visitors (organic traffic)
  - Improves user experience
  - Helps search engines understand content
  - Increases the chances of higher search rankings
- 
- **Key Point:** SEO is the process of optimizing a website so that search engines can crawl, index, understand, and rank it more effectively, making it easier for users to find the website.

## Summary

- HTML5 is the foundation of web development.
- Provides semantic elements and multimedia support.
- Makes websites more accessible.
- Forms and validation improve user experience.
- SEO features help websites rank better in search engines.

**Q1. Which HTML5 semantic element is used to define the main content of a webpage?**

- A) <header>
- B) <main>
- C) <footer>
- D) <aside>

**Answer: B) <main>**

**Q2. Which HTML attribute is used to make a form field mandatory?**

- A) required
- B) mandatory
- C) validate
- D) check

**Answer: A) required**

# Project Based Coding Question – Scenario

You have been hired as a junior web developer to enhance an existing HTML5 course page — the "HTML5 Practical Learning Page" shown in the CODE section that follows.

Your client wants a specific set of improvements made directly inside this single HTML file, using only HTML5 features and without relying on JavaScript unless it is explicitly required. Wherever a change genuinely needs backend or JavaScript support, leave it as a clearly commented placeholder instead.

Your deliverable is the fully updated HTML file, with every task on the following slides implemented directly into the existing structure.

# Tasks – Part 1: Content & Accessibility

**Implement the following directly in the existing HTML5 file:**

1. Add a new Testimonials section between the Media and Embedded Content sections, with at least two student reviews wrapped in `<article>` tags.
2. Insert a “Skip to content” link at the very top of `<body>` that jumps directly to the main content, for accessibility.
3. In the Home section, mark up the word “HTML” so hovering over it shows a tooltip reading “HyperText Markup Language.”
4. Mark up the course start date in a machine-readable format suitable for calendar applications.
5. Add a `:target` selector so whichever section a user navigates to via the nav bar is visually highlighted.
6. Replace the single logo image with a `<picture>` element that serves a smaller image to mobile users while retaining the original for desktop.

**Implement the following directly in the existing HTML5 file:**

1. Add a second `<source>` to the audio element so the file plays correctly in browsers that don't support MP3.
2. Add attributes to the video element so it loops automatically and preloads only its metadata, not the full file.
3. Insert a new Saturday row into the schedule table — “Project Work, 4 Hours, Sarah” — and correctly update the total hours in the table footer.
4. Add scope attributes to all table headers so screen readers can correctly associate them with their rows and columns.
5. Within the registration form, regroup the full name, email, and password fields into their own fieldset labeled “Account Details.”
6. Strengthen the password field using a pattern attribute that requires at least one number and one special character.

# Tasks – Part 3: Form & Final Touches

**Implement the following directly in the existing HTML5 file:**

1. Organize the course dropdown into optgroups labeled “Frontend” and “Backend.”
2. Add a datalist to the personal website field suggesting common domain endings such as .com, .dev, and .io.
3. Modify the FAQ <details> elements so that opening one automatically closes any other that’s open — using only a native HTML5 attribute, not JavaScript.
4. Add code comments explaining why the skills checkboxes cannot be forced to require at least one selection using HTML alone, and what would be needed to enforce that rule.
5. Add code comments describing two realistic ways to fix the form’s broken submission endpoint, which currently results in a 404 error in production.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <!-- =====
    META TAGS
    ===== -->

  <!-- Character encoding -->
  <meta charset="UTF-8">
  <!-- Responsive design -->
  <meta name="viewport" content="width=device-width, initial-
scale=1.0">
  <!-- Basic SEO description -->
  <meta name="description"
  <!-- Keywords - less important for modern search engines, but
useful for practice -->
  <meta name="keywords"
    content="HTML5, HTML tutorial, web development,
forms, tables, semantic HTML">

  <!-- Author -->
  <meta name="author" content="Your Name">

  <!-- Search engine instruction -->
  <meta name="robots" content="index, follow">
```

```
<!-- Page title -->
<title>HTML5 Practical Learning Page</title>

<!-- Favicon -->
<link rel="icon" href="favicon.ico" type="image/x-icon">
</head>

<body>

  <!-- =====
    HEADER
    Semantic HTML Element
    ===== -->

  <header>

    <h1>HTML5 Web Development Course</h1>

    <p>
      Welcome to my <strong>HTML5 practice website</strong>.
      This page demonstrates important HTML concepts.
    </p>

    <!-- =====
      NAVIGATION
      ===== -->

    <nav>
```

```
<a href="#home">Home</a> |
  <a href="#about">About</a> |
  <a href="#topics">Topics</a> |
  <a href="#schedule">Schedule</a> |
  <a href="#media">Media</a> |
  <a href="#contact">Contact</a>
</nav>
</header>
<main>
  <!-- =====
  HOME SECTION
  ===== -->
  <section id="home">
    <h2>Welcome to HTML5</h2>
    <p>
      HTML stands for
      <strong>HyperText Markup Language</strong>.
    </p>
    <p>
      HTML is used to create the
      <em>structure</em> of web pages.
```

```
</p>
<p>
  You can use HTML to create
  <mark>headings</mark>,
  paragraphs, links, images, tables and forms.
</p>
<hr>
<p>
  Example of deleted text:
  <del>Old HTML course</del>
</p>
<p>
  Example of inserted text:
  <ins>New HTML5 course</ins>
</p>
<p>
  Chemical formula:
  H<sub>2</sub>O
</p>
<p>
  Mathematical example:
  x<sup>2</sup> + y<sup>2</sup>
</p>
```

```
</section>
```

```
<!-- =====  
ABOUT SECTION  
===== -->
```

```
<section id="about">
```

```
<h2>About HTML5</h2>
```

```
<article>
```

```
<h3>What is HTML?</h3>
```

```
<p>
```

HTML provides the basic structure of  
a website.

It uses elements such as headings,  
paragraphs,  
images, links and forms.

```
</p>
```

```
</article>
```

```
<article>
```

```
<h3>Why Learn HTML?</h3>
```

```
<p>
```

HTML is the foundation of

**web development**.

It is easy to learn and is used together with  
CSS and JavaScript.

```
</p>
```

```
</article>
```

```
</section>
```

```
<!-- =====  
HEADINGS PRACTICE  
===== -->
```

```
<section>
```

```
<h2>Heading Examples</h2>
```

```
<h1>Heading Level 1</h1>
```

```
<h2>Heading Level 2</h2>
```

```
<h3>Heading Level 3</h3>
```

```
<h4>Heading Level 4</h4>
```

```
<h5>Heading Level 5</h5>
```

```
<h6>Heading Level 6</h6>
```

```
</section>
```

```
<!-- =====
LISTS
===== -->
```

```
<section id="topics">
```

```
<h2>HTML5 Topics</h2>
```

```
<h3>Unordered List</h3>
```

```
<ul>
  <li>HTML Document Structure</li>
  <li>Semantic HTML</li>
  <li>Forms</li>
  <li>Tables</li>
  <li>Multimedia</li>
</ul>
```

```
<h3>Ordered List</h3>
```

```
<ol>
  <li>Learn HTML</li>
  <li>Practice HTML</li>
  <li>Learn CSS</li>
  <li>Learn JavaScript</li>
</ol>
```

```
<h3>Nested List</h3>
```

```
<ol>
```

```
<li>
```

```
  Frontend Development
```

```
<ul>
```

```
<li>HTML</li>
```

```
<li>CSS</li>
```

```
<li>JavaScript</li>
```

```
</ul>
```

```
</li>
```

```
<li>
```

```
  Backend Development
```

```
<ul>
```

```
<li>Node.js</li>
```

```
<li>Python</li>
```

```
<li>PHP</li>
```

```
</ul>
```

```
</li>
```

```
</ol>
```

### <h3>Description List</h3>

<dl>

<dt>HTML</dt>

<dd>Structure of a web page.</dd>

<dt>CSS</dt>

<dd>Used to style a web page.</dd>

<dt>JavaScript</dt>

<dd>Used to add behavior and interactivity.</dd>

</dl>

</section>

```
<!-- =====
HYPERLINKS
===== -->
```

<section>

<h2>Hyperlinks</h2>

<p>

Visit the official

<a href="https://www.w3.org/" target="\_blank">

W3C Website

</a>.

</p>

<p>

Visit

<a href="https://developer.mozilla.org/" target="\_blank">

MDN Web Docs

</a>

to learn more about HTML.

</p>

<p>

<a href="mailto:example@gmail.com">

Send me an Email

</a>

</p>

<p>

<a href="tel:+911234567890">

Call Me

</a>

</p>

</section>

```
<!-- =====
```

## IMAGE

```
===== -->

<section>

  <h2>Images</h2>

  <figure>

    <figcaption>
      HTML5 Logo
    </figcaption>

  </figure>

</section>

<!-- =====
TABLE
===== -->
```

```
<section id="schedule">
```

```
  <h2>HTML5 Course Schedule</h2>
```

```
  <table border="1">
```

```
    <caption>
```

```
      Weekly HTML5 Training Schedule
```

```
    </caption>
```

```
    <thead>
```

```
      <tr>
```

```
        <th>Day</th>
```

```
        <th>Topic</th>
```

```
        <th>Duration</th>
```

```
        <th>Instructor</th>
```

```
      </tr>
```

```
    </thead>
```

```
    <tbody>
```

```
      <tr>
```

```
        <td>Monday</td>
```

```
        <td>HTML Structure</td>
```

```
        <td>2 Hours</td>
```

```
        <td>John</td>
```

```
</tr>
```

```
<tr>
  <td>Tuesday</td>
  <td>Text & Headings</td>
  <td>2 Hours</td>
  <td>John</td>
</tr>
```

```
<tr>
  <td>Wednesday</td>
  <td>Lists & Tables</td>
  <td>2 Hours</td>
  <td>David</td>
</tr>
```

```
<tr>
  <td>Thursday</td>
  <td>Forms</td>
  <td>3 Hours</td>
  <td>David</td>
</tr>
```

```
<tr>
  <td>Friday</td>
  <td>Multimedia</td>
  <td>2 Hours</td>
  <td>Sarah</td>
</tr>
```

```
</tbody>
<tfoot>
  <tr>
    <td colspan="2">
      Total Training
    </td>
    <td colspan="2">
      11 Hours
    </td>
  </tr>
</tfoot>
</table>
</section>
<!-- =====
      AUDIO AND VIDEO
      ===== -->
<section id="media">
  <h2>Audio and Video</h2>
  <h3>Audio</h3>
  <audio controls>
    <source
      src="audio/sample.mp3"
      type="audio/mpeg"
    >
    Your browser does not support audio.
  </audio>
  <h3>Video</h3>
  <video
    width="500"
    height="300"
    controls
    poster="images/video-thumbnail.jpg"
  >
    <source
      src="video/html-course.mp4"
      type="video/mp4"
    >
```

Your browser does not support video.

```
</video>
```

```
</section>
```

```
<!-- =====
EMBEDDING
===== -->
```

```
<section>
```

```
<h2>Embedded Content</h2>
```

```
<h3>YouTube Video</h3>
```

```
<iframe
  width="560"
  height="315"
  src="https://www.youtube.com/embed/UB1O30fR-
EE"
  title="HTML Tutorial"
  allowfullscreen>
</iframe>
```

```
<h3>Embedded Website</h3>
```

```
<iframe
  src="https://example.com"
  width="600"
  height="300"
  title="Example Website">
</iframe>
```

```
</section>
```

```
<!-- =====
FORM
===== -->
```

```
<section id="contact">
```

```
<h2>Student Registration Form</h2>
```

```
<form
  action="/submit-form"
  method="post"
  autocomplete="on"
>
```

```
<!-- Text Input -->
```

```
<p>
```

```
<label for="fullname">
```

Full Name:

```

</label>

<input
  type="text"
  id="fullname"
  name="fullname"
  placeholder="Enter your full name"
  required
  minlength="3"
  maxlength="50"
>

</p>

<!-- Email -->

<p>

  <label for="email">
    Email:
  </label>

  <input
    type="email"
    id="email"
    name="email"
    placeholder="example@gmail.com"

```

required

```

>

</p>

<!-- Password -->

<p>

  <label for="password">
    Password:
  </label>

  <input
    type="password"
    id="password"
    name="password"
    required
    minlength="8"
  >

</p>

<!-- Number -->

<p>

```

```
<label for="age">
    Age:
</label>

<input
    type="number"
    id="age"
    name="age"
    min="10"
    max="100"
    required
>

</p>

<!-- Date -->

<p>

    <label for="dob">
        Date of Birth:
    </label>

    <input
        type="date"
        id="dob"
        name="dob"
    >
```

```
</p>

<!-- Radio Buttons -->

<fieldset>

    <legend>Gender</legend>

    <input
        type="radio"
        id="male"
        name="gender"
        value="male"
        required
    >

    <label for="male">
        Male
    </label>

    <input
        type="radio"
        id="female"
        name="gender"
        value="female"
    >
```

```
<label for="female">
    Female
</label>
```

```
<input
    type="radio"
    id="other"
    name="gender"
    value="other"
>
```

```
<label for="other">
    Other
</label>
```

```
</fieldset>
```

```
<!-- Checkboxes -->
```

```
<fieldset>
```

```
<legend>Select Skills</legend>
```

```
<input
    type="checkbox"
    id="html"
```

```
    name="skills"
    value="html"
>
```

```
<label for="html">
    HTML
</label>
```

```
<input
    type="checkbox"
    id="css"
    name="skills"
    value="css"
>
```

```
<label for="css">
    CSS
</label>
```

```
<input
    type="checkbox"
    id="javascript"
    name="skills"
    value="javascript"
>
```

```
<label for="javascript">
```

```

</label>

</fieldset>

<!-- Select -->

<p>
  <label for="course">
    Choose Course:
  </label>

  <select
    id="course"
    name="course"
    required
  >

    <option value="">
      -- Select Course --
    </option>

    <option value="html">
      HTML5
    </option>

    <option value="javascript">
      JavaScript
    </option>

    <option value="fullstack">
      Full Stack Development
    </option>
  </select>
</p>

<!-- Textarea -->

<p>
  <label for="message">
    Why do you want to learn web development?
  </label>

  <br>

  <textarea
    id="message"
    name="message"
  >

```

```

rows="6"
    cols="50"
    placeholder="Write your answer here..."
    required
    minlength="10"
></textarea>

</p>

<!-- File Upload -->

<p>

    <label for="resume">
        Upload Resume:
    </label>

    <input
        type="file"
        id="resume"
        name="resume"
        accept=".pdf,.doc,.docx"
    >

</p>

<!-- Color -->

```

```

<p>

<label for="color">
    Choose Favorite Color:
</label>

<input
    type="color"
    id="color"
    name="color"
>

</p>

<!-- Range -->

<p>

    <label for="experience">
        HTML Experience:
    </label>

    <input
        type="range"
        id="experience"
        name="experience"
        min="0"

```

```

    max="10"
    value="5" >

</p>

<!-- URL -->

<p>

    <label for="website">
        Personal Website:
    </label>

    <input
        type="url"
        id="website"
        name="website"
        placeholder="https://example.com"
    >

</p>

<!-- Search -->

<p>

```

```

<label for="search">
    Search:
</label>

    <input
        type="search"
        id="search"
        name="search"
        placeholder="Search something..."
    >

</p>

<!-- Hidden Input -->

<input
    type="hidden"
    name="course_id"
    value="HTML101"
>

<!-- Submit and Reset -->

<p>

    <button type="submit">
        Register
    </button>

```

```
</button>
```

```
    <button type="reset">
      Reset Form
    </button>
```

```
</p>
```

```
</form>
```

```
</section>
```

```
<!-- =====
DETAILS / SUMMARY
===== -->
```

```
<section>
```

```
<h2>Frequently Asked Questions</h2>
```

```
<details>
```

```
  <summary>
    What is HTML5?
  </summary>
```

```
<p>
  HTML5 is the modern version of HTML used
```

to structure web pages and applications.

```
</p>
```

```
</details>
```

```
<details>
```

```
  <summary>
    Is HTML a programming language?
  </summary>
```

```
<p>
  No. HTML is a markup language used to
  structure content on the web.
</p>
```

```
</details>
```

```
</section>
```

```
<!-- =====
QUOTATION
===== -->
```

```
<section>
```

```
<h2>Important Quote</h2>
```

```
<blockquote cite="https://www.w3.org/">
    The Web is for everyone.
</blockquote>
</section>

<!-- =====
CONTACT INFORMATION
===== -->

<section>

<h2>Contact Information</h2>

<address>

    Written by Your Name.<br>
    Email:
    <a href="mailto:example@gmail.com">
        example@gmail.com
    </a>
    <br>

    Phone:
    <a href="tel:+911234567890">
```

```
+91 12345 67890
    </a>

</address>

</section>

</main>

<!-- =====
FOOTER
===== -->

<footer>

<hr>

<p>
    &copy; 2026 HTML5 Practical Course.
    All Rights Reserved.
</p>

<p>
    <a href="#home">
        Back to Top
    </a>
</p>
</footer>
</body>
</html>
```